

Understanding the role of satisfaction in the formation of perceived switching value

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ABSTRACT

Satisfaction has long been considered as one critical determinant of the intention to repurchase, continue usage, or switch. Satisfaction is shown to affect intention either directly or via interaction with other factors such as benefits and costs of switching. This study investigates how satisfaction changes individuals' sensitivity toward benefits and costs during switching decision-making. This study incorporates satisfaction into a value-based decision model which originated from the rational decision-making concept in economics and has been widely used to understand the adoption and continuance of innovative products or systems. It is hypothesized that the magnitude of the effects of perceived benefits and costs on the value is contingent on the level of satisfaction in a switching context. Data collected from 237 smartphone users confirm proposed hypotheses that the level of satisfaction determines the importance of benefits and costs in switching value determination. Discussions and implications of this result are provided.

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1. Introduction

Recently, along with the rapid development of mobile telecommunication technologies, the demand for mobile telecommunication services has grown exponentially [9]. Significant academic effort has been made to understand this phenomenon (please see Varnali and Toker [78] for a review). While some researchers focus on the initial adoption of mobile service e.g. [21,40,79], many others pay attention to post-adoption behavior such as the continuance or switching of mobile service [19,22,54,55,69,70,84,85]. In addition to mobile service, one interesting issue in this area is the competition among different smartphone platforms. Smartphone platform refers to the operating system of smartphone, such as Android by Google and iOS by Apple. In addition to Android and iOS, windows phone 8 and Firefox OS joined the battle recently. The market share for Android has grown from 48.7% to 68.4% during 2011–2012, while Apple iOS remains 19.4% [50]. In addition, since the growing trend of overall smartphone market is decreasing, this implies that one important strategy practitioners may take is to encourage platform switching.

In investigating adoption and switching intention regarding innovative products or services, several studies adopted a value-based decision-making research concept to understand the adoption of innovations or switching from one system or service to another [37,39,40,44]. Based on the rational decision-making perspective, theorists argue that the likelihood that rational decision makers will adopt an innovation or change to a different system is high when the adoption or change is found to be worthwhile or valuable. In addition, the value of an innovation is determined by a thorough comparison

between the benefits to be obtained from adoption or switching and the costs to be paid for adopting or switching. In addition to emphasizing the need to take perceived value into consideration, previous study also explored its mediating role between costs, benefits and intention [40]. Drawing on the limitation of including positive effects only in the study (e.g., technology acceptance based model), value-based studies have built a solid foundation for us to better understand the adoption and continuance decision-making process.

Limited by their research contexts (“initial adoption” for Kim et al. [40], and “within an organization” for Kim and Kankanhalli [86]), past related studies considered only cognitive factors (benefits, costs and value) in their models. However, it is understandable that affective factors may also generate certain impacts on the decision-making process. In a non-mandatory switching context, the extent to which individuals are satisfied with the current service or system (affective component) is critical [12]. Switching or continuance intention is a function of cognitive and affective factors and their interaction [34,70]. This highlights the need to take affective components into consideration when a rational decision-making perspective is adopted.

Furthermore, even when affective components were included in the research model, most past studies focused on their direct or interacting effect (with benefits and costs) on the final decision only. The effect of satisfaction on other parts of decision-making is largely ignored. Taking the value-based decision model as an example, different customers may give the same perceived level of benefit or cost a different weight or level of importance when determining the value of taking action. This implies that the magnitude of the effect of benefit and cost on value may be contingent on satisfaction. This also calls into question whether perceived benefits and costs always generate the same effect on perceived value.

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Drawing on the above issues, the purpose of this study is to explore the role of satisfaction on switching decision-making. To better understand the effect of affective components in determining the value of taking action, I move further and explore the effect of satisfaction on other decision-making parts such as the evaluation of switching value (a critical antecedent of switching intention). I articulate the role of satisfaction in value formation based on the needs fulfillment concept found in needs theories. By integrating the value-based decision model and the affective components of switching together, and focusing on the formation of switching value, I attempt to understand the role of satisfaction in different decision-making areas: both the switching value evaluation and the formation of intention. Specifically, I attempt to answer one question: “*Can satisfaction change individuals' sensitivity toward the benefits and costs of switching?*” I refer sensitivity as the magnitude of the effect that benefits and costs generate on the value of switching. I argue that, in addition to affecting intention directly, satisfaction may alter one's sensitivity toward switching benefits and switching costs.

Through clarifying the above issue, this study is expected to contribute to switching research by showing the other role that satisfaction may play in switching decision-making. This study also contributes to the value-based model by showing the need to take affective components into consideration. This study also highlights the fact that the relationship between benefits, costs and value may be contingent on some factors. The rest of this paper is organized as follows. In the second section, I review the value-based decision-making model and the role of satisfaction. Corresponding hypotheses are then built. The third section describes the method used to collect the required data. The fourth section includes hypothesis testing and discussion. Implications and conclusions are included in the last section.

2. Literature review and hypothesis development

2.1. Value-based decision model

Based on the cost–benefit analysis concept, Kim et al. [40] proposed a value-based adoption mode to understand the determinants of M-commerce adoption decision-making. They argued that the traditional technology acceptance model (TAM), which is widely adopted to understand employees' acceptance of new technologies (e.g., a spreadsheet or word processor) in an organizational setting, is insufficient because the TAM takes only the benefits of innovations into consideration. Different from information systems users in organizations, consumers of contemporary information and communication technologies should be viewed as customers in many circumstances. As a result, costs should not be neglected since customers have to pay for the innovations and adapt their usage behaviors to fully utilize the innovations. Kim et al. [40] further proposed that, in addition to taking costs into consideration, customers determine the net value by making a thorough comparison between the benefits (e.g., usefulness and enjoyment) and costs (e.g., technicality and perceived fee) of adoption. As a result, net value mediates the effects of benefits and costs and serves as a major determinant of adoption intention.

Subsequent researchers adopted this concept and examined the importance of perceived value in the intention to pay for online auctions [46], banking service [1,3] and many other services. Meanwhile, some studies incorporated a multidimensional value concept and explored the antecedents of different dimensions of value [62]. While some studies adopted the value concept but focused solely on the impact of benefits on perceived value [44], other studies emphasized only the effect of perceived cost on perceived value. For example, Kim and his colleagues studied the differences between potential and repeat customers regarding the effect of perceived cost on perceived value [41]. Kim et al. [42] compared the importance of trust and price to perceived value and purchase intention for both potential and repeat customers [42]. In addition, although the proposed value-based adoption

model was developed to understand the initial adoption intention, the following studies adopted this model and applied it in different research contexts. For example, Lin et al. [45] proposed a value-based continuance intention model which integrated the value concepts and traditional expectation–disconfirmation theory.

Recently, in addition to being used in studies of initial adoption and continuance usage, the value-based model has also been used to understand the intention to switch or change. Ko et al. [44] developed a value-based model to understand why an organization's users resist changing to innovative information systems. Based on status quo bias theory and the equity implementation model, they hypothesized and found that switching benefits have a positive impact and switching costs have a negative impact on perceived value. Users are less likely to resist the change if they find it valuable. In addition, the researchers also found a strong direct effect from switching costs to resistance. In subsequent research, Kim [39] further classified switching costs into *uncertainty*, *transition*, *sunk*, and *loss costs* and examined both their direct and indirect effects (through perceived value) on resistance to change. The result shows that, while switching benefits have a strong effect on perceived value, transition and loss costs significantly affect the perceived value of change.

As indicated above, perceived value is critical to the switching decision-making process. Previous studies also pointed out that perceived value is determined by both benefits and costs. Although previous studies built a solid foundation for understanding the importance of value in decision-making, some interesting questions remain unanswered. For example, the research context for Ko et al. [44] and Kim [39] was an organizational setting in which usage is most likely mandatory. In addition they took the rational decision-making perspective only and considered solely benefits and costs. However, it is reasonable to understand that affective components also play a role in the decision-making or attitude-formation process, especially in a non-mandatory use continuance (or switching) context. In fact, marketing literature has pointed out that satisfaction is one of the critical drivers of switching behavior. For example, Bansal et al. [12] viewed switching as one type of migration and applied the popular research model named push–pull–mooring (PPM) to classify major drivers of and barriers to switching. *Pushing* drivers contain factors (such as dissatisfaction) that encourage an individual to leave current choice. *Pulling* drivers (such as alternative attractiveness) represent a new choice's attractiveness to individuals. *Mooring barrier* refers to factors such as switching cost which inhibit individuals from switching from their current choice to a new one. Bansal et al. [12] articulated pull, push and mooring as three second order factors and successfully illustrated their effects on switching intention. This model draws significant attention in the marketing and IS area. Although different factors were used to represent pull, push and mooring in different contexts [12,31,32,83], they all included satisfaction in the pushing construct.

2.2. The role of satisfaction: a needs theory perspective

The literature review hints at a need to take satisfaction, an affective component, into consideration. Satisfaction, indeed, is an important antecedent of the continuance to use innovative information technologies. Among various published studies, one important research stream regarding satisfaction is *needs theory* [6]. Needs theories highlight the importance of needs fulfillment [2,29,48,51]. This research stream argues that individuals are less satisfied when some of their needs are not fulfilled and, in contrast, they are more content if the service or system provided fulfills their needs. As an outcome, satisfaction shows certain level of correlation with needs fulfillment. In fact, some IS studies even incorporate needs fulfillment into satisfaction measurement [8,28]. The above discussion hints at a need to focus on needs fulfillment while attempting to understand satisfaction. Indeed, the importance of needs fulfillment has been highlighted in one recent study [7]. Equitable

needs fulfillment is found to have a significant impact on satisfaction after taking the effect of IS performance into consideration.

In addition to the high correlation between needs fulfillment and satisfaction, another important assumption of need theories is that unsatisfied individuals are motivated to take action to remove deficiencies caused by a need [73]. Individuals have stronger desires when their needs are less fulfilled [2] because unfulfilled needs make individuals to feel not only upset but also uncomfortable [49]. Low satisfaction can be viewed as an emotional state that prompts customers to evaluate alternative courses of action to reduce the existing dissatisfaction state and/or to attain a future satisfaction state [72]. In sum, low satisfaction caused by unfulfilled needs motivates customers to take action to have their needs fulfilled to ease the feeling of discomfort.

Switching is one possible action that customers may take if current service or system cannot satisfy them. Satisfaction does play a critical role in determining the switching intention given that a fair number of empirical switching studies incorporate satisfaction into their research models [23,38]. It is noticeable, however, that even though most studies include satisfaction in the model, many of them hypothesized a direct effect from satisfaction to switching intention. The decision to switch is made after the initial adoption decision. The actual usage process generates affective outcomes, e.g., satisfaction, and those affective outcomes, without doubt, are critical for continuance decision-making. For example, early expectation–disconfirmation studies indicate that satisfaction is one critical determinant, if not the most important one, of continuance intention [56,59,61]. Later on, some studies moved further and hypothesized that, in addition to influencing intention directly, satisfaction may interact with other factors in the decision-making process. For example, continuance intention is also determined by the interactions between satisfaction and switching cost and between satisfaction and alternative attractiveness [2,10,70]. Those later studies concluded that although users who are more satisfied are more likely to continue their current service, the costs and benefits of switching could either strengthen or weaken this relationship.

As indicated above, although the effect of satisfaction has been examined, most past studies quested for its direct or interacting (with other factors) effect on the final intention only. However, it is reasonable to suspect that satisfaction may have an effect on other parts of decision-making. Taking the value-based decision model as an example: while satisfaction reduces switching intention, it might also change value evaluation. By viewing satisfaction as an outcome of needs fulfillment, satisfied individuals can be viewed as those whose needs are fulfilled and dissatisfied individuals as those whose needs are not fulfilled. For those satisfied individuals, given that their needs are fulfilled, their motivation to leave current service provider or abandon current system is then low. Since their needs have been fulfilled, benefits provided by other service providers or systems are not so valuable to them. In addition, since switching is not necessary (to have their needs fulfilled), the cost of switching significantly reduces the value of switching given that the cost is considered as extra and unwanted. In contrast, for those unsatisfied individuals, their needs are not fulfilled and their motivations to leave current service provider are much stronger. Those eager-to-leave individuals tend to rate benefits that cannot be obtained from their current service providers or systems as being more valuable. Meanwhile, they also tend to consider the cost of switching to be necessary to the fulfillment of their needs. As an outcome, the value of switching does not decrease significantly as cost increases. Therefore, it is reasonable to suspect that individuals with different levels of satisfaction with their current service or system may perceive the value of switching differently even though they sense the same level of benefits or costs. This phenomenon is considered as “satisfaction changing the level of value sensitivity.” Value sensitivity is defined as the magnitude of the impact that benefit and cost generate on value. It represents the magnitude of the value change on each unit change of benefits or cost. Higher sensitivity reflects on a

larger absolute value of the coefficient of the links between benefit, cost and value. In sum, this study argues that satisfaction also plays a role in the value formation stage by changing value sensitivity, in addition to generating an effect on switching intention directly.

A research model is developed based on the above arguments. As shown in Fig. 1, based on the value-based decision model, perceived switching value is a function of perceived switching benefit and perceived switching cost. In addition, this study incorporates satisfaction and explores its role in different decision areas. The newly added affective component, *satisfaction*, is expected to reduce switching intention and moderate the effect of both perceived switching benefit and cost on perceived value. In the following, corresponding arguments for each link are provided.

2.3. Hypothesis development

This hypothesis development section includes two major parts. The first part focuses on the value-based decision model and the building of links among switching benefit, cost, value and intention. The second part concentrates on the role of satisfaction, including its direct effect on switching intention and the moderating effects of the relationships from benefit and cost to value.

Perceived switching benefit is the major pulling factor which drives individuals to move across different service providers [12]. *Pulling factors* are “positive factors drawing prospective migrants to the destination” [53]. Bansal et al. [12] also pointed out that alternative attractiveness is the only existing variable specified in switching literature. *Alternative attractiveness* refers to the positive characteristics of competing service providers which positively increases customers' intention to switch [34]. The following switching literature, based on the push–pull–mooring model, used relative advantage or attractive alternatives to represent the pulling factor of switching e.g. [32,82,83]. Alternative attractiveness is shown to have a positive impact on switching. Jones et al. [34] argued that when individuals find no better alternatives, higher levels of retention can be observed. In fact, alternative attractiveness has also been used to explain employee turnover [24] career commitment [26], channel relationships [66], online games [32]; social networking sites [82], and blogs [83].

On the other hand, in their study of resistance to change, Ko et al. [44] defined *switching benefit* as the perceived utility a user would enjoy in switching from the status quo to the new information system. Switching benefit is specifically viewed as the increase in outcomes and the decrease in inputs. Since our research context is mobile platform, the benefit of switching to other mobile platforms is defined as a customer's enjoyment of switching from the status quo to the new service provider. Although previous studies in this research stream e.g. [39,44] did not hypothesize the direct effect from switching benefit to switching intention, a positive relationship between these two variable can be expected, based on alternative attractiveness studies. When customers find that an alternative platform is of better quality, they find less need to stay with their platform. Therefore, I hypothesize:

H1. Perceived switching benefit is positively associated with switching intention.

Perceived switching cost refers to customers' perceptions of the time, money and effort associated with changing platform [34]. It is the perceived disutility a user would incur in switching [44]. Recent studies found perceived switching cost to be a significant determinant for the intention of behaviors in contexts such as mobile services [38], wireless financial services [43], mobile banking [47] and mobile commerce transactions [80]. In the PPM model, switching costs belong to the mooring factor which serves as barrier to switching [12]. Several studies attempted to classify barriers to switching into different dimensions or categories e.g. [16,30,34]. Even though different categorizations can be found based on different perspectives, theorists all agree that switching cost is a useful predictor of customer retention and switching

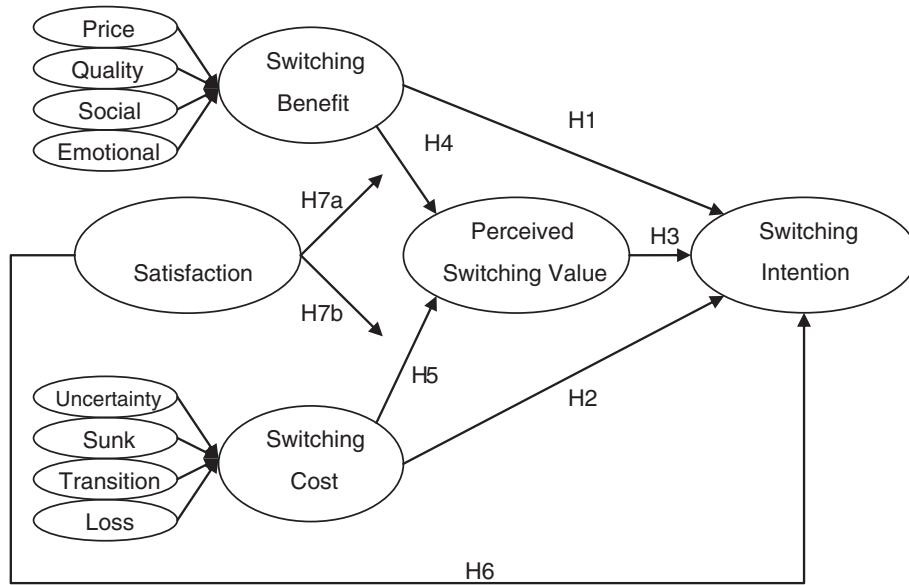


Fig. 1. Research model.

[4,12,16,32]. Customers or users tend to stay with their current service provider or system when the switching cost is high. For example, they tend to avoid losses associated with change, avoid uncertainty resulting from change, and do not want to forgo their past investments in their current service. In a mobile context, various costs associated with switching platforms reduce customers' intention to take action. Those costs include extra spending on obtaining new devices, the perceived loss of past investments in equipment or software, or the increased time and effort required to learn a new service. Following this research stream, this study also builds the link between switching cost and switching intention and hypothesizes the following:

H2. Perceived switching cost is negatively associated with switching intention.

2.3.1. The role of perceived switching value

In addition to the direct effect from perceived benefit to switching intention, aligning with the literature, I also hypothesize that perceived switching benefit promotes switching intention indirectly through increasing the perceived switching value. I first build the link between perceived switching value and switching intention, and then the effects from perceived switching cost and benefit to perceived value.

Perceived value is the perceived net benefit of performing a behavior [40]. Perceived value is closely associated with behavioral intention e.g. [15,52,75]. In the research context of switching, the perceived value of switching refers to perceived benefits relative to the costs of changing platforms. The net benefit that can be obtained from change is negatively associated with resistance to change [44,68]. Since people have a strong tendency to maximize value in their decision-making [71], change or switching is favored when users or customers find that the net gain from switching providers is positive [35].

Since value is what customers gain over what they lose in terms of switching, a *positive* relationship between gain and value and a *negative* relationship between loss and value are then expected. For example, when facing the same level of cost, higher benefit represents more net value that customers can receive. On the other hand, higher switching costs would decrease the users' net benefits (or their perceived value) of the change because net benefit is assessed by weighing benefits relative to the costs of change. Therefore, I follow past studies and hypothesize the following:

H3. Perceived switching value is positively associated with switching intention.

H4. Perceived switching benefit is positively associated with perceived switching value.

H5. Perceived switching cost is negatively associated with perceived switching value.

2.3.2. The role of satisfaction

As indicated by expectation–disconfirmation theory [58], satisfaction is the most critical factor in determining the intention to continue current service. In a switching context, the negative relationship between satisfaction and switching intention has been well articulated [11]. Theorists argue that customers are satisfied when their current service provider (or platform in this study) is able to meet their expectations and, therefore, satisfied customers are more likely to stay with their current platform [60,81]. As a result, the willingness of satisfied customers to switch to another platform is low. In contrast, dissatisfied customers are those whose expectations or needs are not fulfilled by their current platform. In order to have their needs fulfilled, they tend to switch to another platform. Therefore, satisfaction/dissatisfaction is classified as one pushing factor which drives individuals to leave their current platform [12]. Empirical studies also demonstrated a strong correlation between satisfaction and switching intention [17,20,32]. Therefore, the following hypothesize is then made:

H6. Satisfaction is negatively associated with switching intention.

In the previous section, I hypothesized that perceived benefit has a positive main effect on perceived switching value. When they find that an alternative platform provides more benefits than their current one, customers are more likely to determine that switching is valuable. In this section, I further argue that this relationship is contingent upon the level of satisfaction. As indicated above, the level of satisfaction is determined by the extent to which customers' needs are fulfilled and expectations are met by their current platform. Satisfied customers are less likely to leave their current platform because their needs are fulfilled and, therefore, there is no force driving them to leave [12]. Given that their needs are fulfilled, extra benefits provided by other service providers are then not so attractive since switching is not without cost. In addition, status quo bias studies indicate that people in general

have stronger status quo bias when their current platform is functioning well [77]. Therefore, satisfied customers tend to believe that switching is less valuable. It is reasonable to expect that the coefficient between perceived benefits and perceived value will be lower for satisfied individuals.

One the other hand, unsatisfied customers will be more sensitive to the benefits of switching is expected. *Sensitivity* is the magnitude of the effect that the switching benefit has on switching value. Unsatisfied customers are those whose needs are not fulfilled by their current platform and are motivated to take action to have their needs fulfilled. *Switching benefit* refers to positive outcomes that can be received after switching to an alternative. In general, those positive outcomes include needed features not offered by the current service provider. Given that a low satisfaction level is caused by deficient fulfillment of needs, having those unfulfilled needs fulfilled is then important and highly valued. Therefore, unsatisfied customers tend to believe that switching to another platform is worthwhile and a wise decision. The following hypothesize is then made:

H7a. Compared with satisfied individuals, the effect of perceived switching benefit on perceived switching value is stronger for unsatisfied individuals.

As indicated, perceived switching cost is expected to be negatively associated with perceived switching value. High switching costs drive individuals to believe that switching is not valuable and, therefore, block them from leaving their current platform. In this section, I further argue that, when determining the value of switching, highly satisfied individuals tend to be more sensitive to the level of switching cost. Status quo bias can be observed in general switching decision-making [44,68]. According to Kahneman and Tversky [36], losses loom larger than gains when people perceive value. This is especially true for highly satisfied individuals since there is no need for them to leave their current platform. Highly satisfied individuals are those whose needs or expectations are fulfilled by their current platform. Given that they are not motivated to leave, any tiny switching cost is considered pure loss which should be avoided. They are therefore more sensitive to switching cost. For example, the extra learning cost and uncertainty about the future lead highly satisfied individuals to believe that sticking with their current platform is a better choice and, therefore, the value of switching is low.

In contrast, dissatisfied individuals are more eager to move to another place that can fulfill their needs. Since their goal is to have their needs fulfilled, they pay more attention to how well the goal is reached but not to the potential costs of achieving it. As long as those unsatisfied customers believe that other platforms can better fulfill their needs or meet their expectations, they tend to rate the value of switching high even though its cost is high. For example, customers may have to give up past investments and learn how to use a new smartphone platform after switching. When facing the same level of switching cost, unsatisfied customers tend to believe that the perceived cost of giving up past investments is not that high and spending time and effort on learning is minor, if the prospective service provider can fulfill their needs. Hence, unsatisfied customers are less sensitive to switching cost is then hypothesized.

H7b. Compared with unsatisfied individuals, the effect of perceived switching cost on perceived switching value is stronger for satisfied individuals.

3. Research method

3.1. Data collection

Based on the purpose of this study, opinions from mobile phone users are collected to examine the proposed model. The research target of this study is customers with smartphone experience in Taiwan. The items of the questionnaire were adapted from related studies published in academic journals. In addition, since the data collection took place in Taiwan, a translation and back-translation approach was adopted to ensure the quality of the translation. The author of this study first translated the English version questionnaire into Chinese. A translator unaware of the research content was asked to translate the Chinese version back to English. A thorough comparison was made to check the consistency of the two English versions and no significant differences were found. To ensure content validity, a pre-test and a pilot test were conducted to evaluate the quality of the questionnaire. Along with 7 Ph.D. students, 3 professors with experience in a related research area were asked to read each item and figure out possible ambiguities. In addition, the 7 Ph.D. students also performed a card-sorting task. They were asked to categorize all used items into a number of categories based on the perceived semantic meanings. Items located in different constructs were modified or removed before conducting the pilot test. In the pilot test, a total of 88 participants with smartphone experience were selected as subjects. Data pertaining to factor and reliability analysis were used. In this stage, only one unclear item in the demographic section was modified.

3.2. Sample

The official survey was conducted in Taiwan. I posted the survey on a popular online smartphone forum. A total of 255 samples were collected during February 2012. Among those, 237 were completed and considered valid responses. Based on the system log of that online forum, about 800 registered members accessed the forum during that period; the estimated response rate is then 29.63%. This response rate is attributed to the gift coupon lottery, provided as an incentive. Detailed information about the sample of this study is provided in Table 1.

3.2.1. Construct and measurement

The survey measures for the study were derived from previous published studies. *Switching intention* was measured with 3 items adapted from Kim et al. [40]. These three items aimed at capturing the extent to which respondents would switch to another platform in the near future. For *switching value*, three items adapted from [40,71] were used to capture the extent to which benefits to be received from switching were worthwhile, in terms of the sacrifices required. For *satisfaction*, three items adapted from [12,57] were used to capture the

Table 1
Demographic information.

Measure	Categories	#	%	Measure	Categories	#	%
Switching experience	Yes	126	53.2	IT knowledge	Very low	2	0.8
	No	111	46.8		Low	15	6.3
Gender	Male	114	48.1	Educational background	Fair	99	41.8
	Female	123	51.9		High	91	38.4
The same platform used by people surrounding	Very low	9	3.8		Very high	30	12.7
	Low	22	9.3		High school	3	1.3
	Half	117	49.4		College	126	53.2
	High	66	27.8		Graduate	108	45.6
	Very high	23	9.7				

Note: N = 237.

extent to which customers were delighted, satisfied and content with their current platform.

Given the multi-dimensional nature of switching benefits and costs, I operationalized them as second order formative constructs, each of which contained four first order reflective constructs [65]. *Switching benefits* refers to the extent to which more benefits can be received from alternative platforms. Constructs and items to measure switching benefits were adapted from Sweeney and Soutar [74]. They classified the value that services or products may offer into four dimensions: emotional, social, quality and price. The emotional dimension aims at capturing the enjoyment or pleasure derived from the alternative. The social dimension measures the social consequences of what the alternative communicates to other people. The quality dimension includes customers' perceptions of the reliability and durability of the alternative. The price dimension focuses on the relative price of alternatives.

Constructs and items for *switching costs* were adapted from [16]. Among those costs identified by previous studies, four that were most related to the research context of this study were selected: monetary loss costs, benefit loss costs, set-up costs and learning costs. Monetary loss cost captures the onetime financial outlay that is incurred in switching to another platform. Benefit loss costs are the costs associated with contractual linkages that create economic benefits for staying with an incumbent platform. Set-up costs are the time and effort costs associated with the process of setting up for initial use. Learning costs are the time and effort costs of acquiring new skills or know-how in order to use the new platform. Each item in this survey used a Likert-type scale ranging from 1 (completely disagree) to 5 (completely agree) to express the opinion of each respondent. Furthermore, this study used all

positive response items to measure some constructs, such as switching intention and switching value, because positive response only approach is widely adopted by past studies e.g. [14]. The actual items used in the survey are presented in Table 2.

Tables 2 and 3 also include validity and reliability information. In this study, partial least squares (PLS) analysis was used to test item reliability, convergent validity, and discriminant validity [33]. Individual item reliability can be examined by observing the factor loading of each item. High loading represents that the shared variance between constructs and its measurement is higher than error variance. Factor loading higher than 0.7 can be viewed as high reliability and items should be dropped when factor loadings are less than 0.5. In addition, item-total correlations (ITCs) are the representativeness of items according to the scale and whether the scale is reliable. As shown in Table 2, all factor loadings are significant and higher than the recommended 0.7. The item-total correlations are also higher than the recommended cut-off value of 0.3. Therefore, individual item reliability is ensured in this study.

Convergent validity should also be examined when two indicators are used to measure the constructs. Convergent validity was examined with factor loadings, the composite reliability (CR) of constructs and average variance extracted (AVE) by constructs [25]. High factor loading represents that items are highly correlated with the construct, CR reflects the extent to which items within one construct are consistent, and AVE represents the common part of all indicators within one construct. Convergent validity is ensured since factor loadings are higher than 0.7, composite reliability of all constructs is higher than 0.7, and the AVE values are all greater than 0.5. Finally, as shown in Table 3 the

Table 2
Constructs, measurements, and factor loadings.

Constructs	Items	Loadings	ITC
Switching intention CR: 0.95; AVE: 0.87; Alpha: 0.93	I am considering switching from my current <i>Smartphone platform</i> soon.	0.91*	0.82
	The likelihood of me switching to another <i>Smartphone platform</i> is high.	0.95*	0.89
	I am determined to switch to another <i>Smartphone platform</i> .	0.94*	0.85
Perceived switching value CR: 0.96; AVE: 0.89; Alpha: 0.94	Considering the time and effort that I have to spend, the change to the new <i>Smartphone platform</i> is worthwhile	0.94*	0.87
	Considering the loss that I incur, the change to the new <i>Smartphone platform</i> is of good value	0.94*	0.86
	Considering the hassle that I have to experience, the change to the new <i>Smartphone platform</i> is beneficial to me	0.95*	0.89
Satisfaction CR: 0.95; AVE: 0.87; Alpha: 0.92	Very dissatisfied Very satisfied	0.92*	0.82
	Very displeased Very pleased	0.95*	0.88
	Terrible Delighted	0.92*	0.82
Switching benefit Emotional CR: 0.96; AVE: 0.83; Alpha: 0.95	<i>Alternative platform</i> is the one that I enjoy more	0.92*	0.83
	... makes me want to use them more	0.92*	0.81
	... is one that I feel more relaxed about using	0.88*	0.82
	... makes me feel better	0.91*	0.92
	... gives me more pleasure	0.91*	0.92
Switching benefit Price CR: 0.88; AVE: 0.66; Alpha: 0.84	<i>Alternative platform</i> is more reasonably priced	0.67*	0.88
	... offers more value for money	0.83*	0.91
	... is good relative to the price more	0.91*	0.91
	... is more economical	0.85*	0.62
Switching benefit Quality CR: 0.94; AVE: 0.80; Alpha: 0.91	<i>Alternative platform</i> has more consistent quality	0.88*	0.83
	... is better designed	0.89*	0.91
	... has a higher acceptable standard of quality	0.88*	0.85
	... has better operationalizability (ease of use)	0.91*	0.88
Switching benefit Social CR: 0.96; AVE: 0.85; Alpha: 0.94	<i>Alternative platform</i> helps me feel more acceptable	0.88*	0.89
	... improves the way I am perceived	0.94*	0.88
	... makes a better impression on my friends	0.94*	0.91
	... gives me higher social approval	0.91*	0.88
Switching cost – Uncertainty CR: 0.90; AVE: 0.69; Alpha: 0.85	Switching to a new platform will probably involve hidden costs/charges.	0.79*	0.84
	I am likely to end up with a bad deal financially if I switch to a platform.	0.82*	0.88
	Switching to a new platform will probably result in some unexpected hassle.	0.84*	0.87
	I don't know what I'll end up having to deal with while switching to a platform.	0.86*	0.79
Switching cost – Loss CR: 0.90; AVE: 0.75; Alpha: 0.83	Switching to a new platform would mean that I have to abandon those applications that I have paid.	0.84*	0.82
	How much would you lose in downloaded applications if you switched to a new platform? (lose nothing..., lose a lot)	0.88*	0.84
	I will lose benefits (e.g. iCloud) of being a customer if I leave current platform.	0.87*	0.86
	Switching to a new platform would involve some up-front costs (set-up fees, membership fees, deposits, etc.).	0.79*	0.79
Switching cost – Sunk (MLC) CR: 0.90; AVE: 0.76; Alpha: 0.84	How much money would it take to pay for all of the costs associated with switching platform? (no money..., a lot of money)	0.92*	0.92
	Switching to a new platform costs me a lot of money	0.90*	0.90
	It takes time to go through the steps of switching to a new platform.	0.92*	0.92
	Switching platform involves an unpleasant process.	0.92*	0.92
Switching cost – Transition (SC) CR: 0.94; AVE: 0.83; Alpha: 0.90	There are a lot of formalities involved in switching to a new platform.	0.89*	0.89

* $p < 0.01$.

Table 3
Descriptive analysis and correlation matrix.

Variables	Mean	Std	M3	M4	Correlation matrix										
					SB	EB	PB	QB	LC	UC	MC	SC	Sat	SV	SI
Social benefit	3.18	0.87	0.03	−0.11	0.92										
Emotional benefit	3.44	0.87	−0.24	−0.41	0.54	0.91									
Price benefit	2.91	0.81	−0.21	−0.27	0.31	0.52	0.81								
Quality benefit	3.46	0.90	−0.38	−0.53	0.49	0.75	0.46	0.89							
Loss cost	3.84	0.66	−0.23	0.01	0.01	−0.11	−0.06	−0.05	0.87						
Uncertainty cost	3.81	0.66	−0.54	0.49	0.01	−0.12	−0.10	−0.11	0.44	0.83					
Monetary cost	4.01	0.64	−0.28	0.03	0.13	0.01	−0.09	0.06	0.42	0.42	0.91				
Set-up cost	3.27	0.80	−0.46	−0.45	0.04	−0.06	0.03	−0.02	0.21	0.48	0.31	0.87			
Satisfaction	3.86	0.77	−0.42	0.02	−0.16	−0.18	−0.16	−0.19	0.31	0.24	0.14	0.11	0.93		
Switching value	2.94	0.86	0.05	−0.80	0.30	0.41	0.39	0.42	−0.17	−0.24	−0.16	−0.23	−0.27	0.94	
Switching intention	2.72	1.06	0.11	−0.75	0.33	0.44	0.33	0.42	−0.16	−0.16	−0.03	−0.06	−0.40	0.52	0.93

Note: *Square root of average variance extracted (AVE) appears along diagonal; **M3: Skewness; M4: Kurtosis.

correlations between pairs of constructs are below 0.6 and the square root of AVE is higher than each corresponding correlation coefficient [18]. These properties indicate that the construct measures are sufficiently distinct from each other. Therefore, discriminant validity is also ensured.

For the two second order formative constructs, although each first order construct was measured with reflective indicators, and a series of measures were used to assess the quality of measurement, it was still necessary to examine the appropriateness of treating behavioral integration as a second order formative construct. The procedures proposed by Pavlou & El Sawy [63] are followed to verify the existence of a second order formative construct. First, the relative weights of the first order constructs are all significant. Second, the moderate level of correlation coefficients among variables indicates that a reflective model seems less likely. Finally, low VIF values (all less than 3.3) indicate that these first order dimensions represent different meanings and should not be treated as reflective.

3.3. Common method bias

Common method bias should be examined before performing hypothesis testing when both independent and dependent variables are collected simultaneously from the same respondent. Two tests were conducted to examine common method bias in the collected data. First, I performed a Harman's single factor test with all indicators

used in this study. A total of 10 factors were extracted, and the first factor explained 28% of the variance. Second, in PLS, one method variable was created and linked to both independent and dependent variables [64,67]. Almost all loadings of this method variable were found to be insignificant. Therefore, common method bias should not be a problem in this study, based on the above evidences.

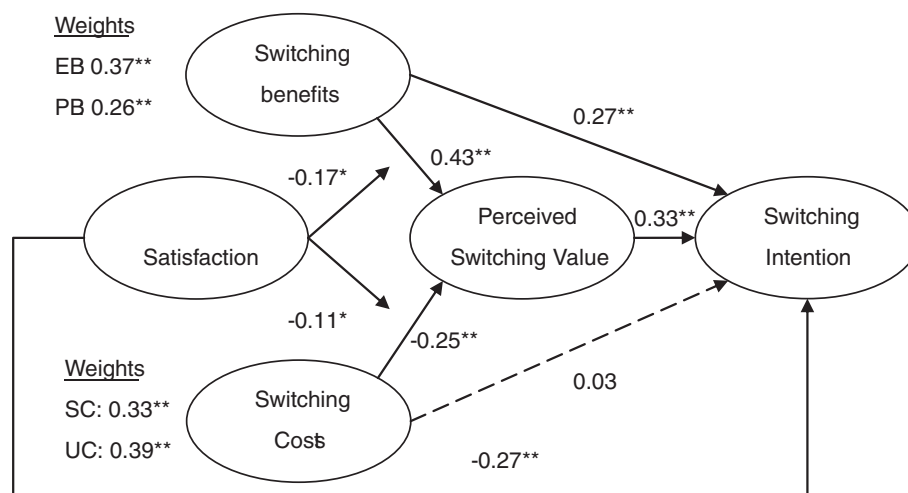
3.4. Non-response bias

In order to detect the potential bias resulting from sampling, a comparison of the early and late respondents was conducted on all variables (with the late respondents being assumed to be similar to non-respondents) [5]. The results show no significant differences between these two groups in all constructs. Therefore, the credibility of the following analysis is not undermined by non-response bias.

4. Data analysis

4.1. Hypothesis testing

PLS was used to test the proposed hypotheses. The bootstrapping technique was used with a bootstrap sample of 500 to evaluate the significance level of the proposed links. Test results are shown in Fig. 2. For the valued-based decision part in the model, first, perceived switching benefits ($\beta = 0.27$; t -value = 3.89) and perceived switching value



EB: emotional; PB: price; QB: quality; SB: social
SC: Sunk; UC: uncertainty; LC: loss; TC: transition

*: $p < 0.05$; **: $p < 0.01$

Fig. 2. Path analysis.

($\beta = 0.33$; t -value = 4.43) were found to have positive effects on switching intention. However, the link between perceived switching cost and switching intention is insignificant ($\beta = 0.03$; t -value = 0.94). Therefore, H1 and H3 were supported but H2 was not. Second, while switching benefit ($\beta = 0.43$; t -value = 7.77) was found to have a positive impact on perceived switching value, perceived switching cost ($\beta = -0.25$; t -value = 4.25) affected perceived switching value negatively. Therefore, both H4 and H5 were supported. The affective component (satisfaction) was found to reduce switching intention ($\beta = -0.27$; t -value = 4.65). In order to test the moderating effect, two interaction terms were generated. Since benefits and costs contain many items, the application of product indicator approach directly reduces the statistical power significantly under this condition [27]. An averaged score of each benefit and cost component was calculated before conducting product indicators [76]. Satisfaction was found to moderate the effects of switching benefit ($\beta = -0.17$; t -value = 2.39) and switching cost ($\beta = -0.11$; t -value = 1.99) on perceived switching value. Therefore, H6, H7a, and H7b were supported.

The test results also showed that 40.3% of the variance of switching intention is explained by perceived switching benefit, perceived switching cost, perceived switching value, and satisfaction. In addition, perceived switching benefit, switching cost, satisfaction, and their interaction explain more than 35% of the variance of perceived switching value.

4.2. Discussion

There are some differences between this study and previous studies e.g. [39,44]. First, this study follows the push–pull–mooring research model and hypothesized a link between perceived switching benefit and switching intention which is absent in previous value-based decision models. In fact, even though this link was not hypothesized in previous studies, a moderate to strong correlation between perceived switching benefit and behavioral intention can still be observed (e.g., 0.59 in Kim [39]). The found results show that perceived switching benefit has a strong and positive effect on switching intention. In addition, the effect of perceived switching benefit on perceived switching value and the effect of perceived switching value on switching intention were all found to be significant. This indicates that perceived switching value only partially mediates the effect of perceived benefit on intention [13].

One difference between the found results and those of previous studies is that switching cost has no direct effect on switching intention. However, certain level correlation (-0.19) between switching cost and intention can be observed in Table 3. The coefficient from switching cost to switching intention is significant when switching intention is regressed with switching benefit and cost exclusively ($\beta = -0.11$; t -value = 1.98). This link from switching cost to intention becomes insignificant after linking perceived switching value to intention. Given that the links from switching cost to switching value and from switching value to switching intention are all significant, the conclusion that the effect from cost to intention is fully mediated by perceived switching value can then be reached [13].

Although the relationship from satisfaction and switching intention was found to be significant, the strength of the relationship is only moderate. This result is similar to other switching based studies; satisfaction toward current product or service only has a low to moderate level impact on switching intention e.g. [12]. However, it is different from past continuance-based studies that found a strong relationship between satisfaction and repurchase intention e.g. [34]. It is understandable that satisfaction is a major driver of repurchase or continuance intention in expectation–disconfirmation or related marketing theories. However, its effect is relatively minor since customers also consider other factors such as the benefits and costs of switching when making a switching decision. Therefore, the effect of satisfaction is not as strong here as it is in non-switching theories.

Second, although I investigated smartphone platform switching intention based on a value-based decision model, I took satisfaction into consideration and hypothesized its effect on switching intention. Many previous studies ignored this link, given that their research context was mandatory usage in which satisfaction may not have a significant effect. However, I believe, as many other studies do, that satisfaction should be taken into consideration in a voluntary usage or purchasing context. The results also showed that satisfaction with the current service provider has a negative impact switching intention.

Third, in addition to the direct link from satisfaction to switching intention, I also hypothesized the moderating effect of satisfaction. As shown in Fig. 2, satisfaction negatively moderates the effects of benefit and cost on value. For the links from benefit to value, the negative moderating effect implies that more satisfied individuals are less sensitive to the benefits since the main effect from benefit to value is positive. On the other hand, with the negative main effect from cost to value, the negative moderating effect of satisfaction reflects that more satisfied individuals are more sensitive to the effects of switching cost.

In the following, two diagrams were created to further illustrate the above concepts. I first separated the sample into three groups based on the level of satisfaction (low, medium, and high satisfaction). I then regressed perceived switching value with switching cost individually for different satisfaction groups. The three sets of coefficients obtained, including constant and beat, were then used to create Fig. 3a. The same procedures were used to create Fig. 3b, with perceived switching benefit as the independent variable.

The R-square values for three different groups are 0.310 for low satisfaction, 0.237 for medium satisfaction, and 0.100 for high satisfaction. This implies that perceived benefit is relatively important to perceive value of switching when the level of satisfaction is low. As shown in Fig. 3a, the deeper slope of the low satisfaction group indicates that those individuals are more sensitive to the benefits of switching. As the perceived switching benefit increases, dissatisfied individuals tend to believe that switching is more valuable than highly satisfied individuals do. In addition, when the perceived switching benefit is low, both the high and low satisfaction groups perceive the value of switching as extremely low. However, when perceived switching benefit is high, less satisfied customers tend to believe the value of switching is much higher than highly satisfied customers do.

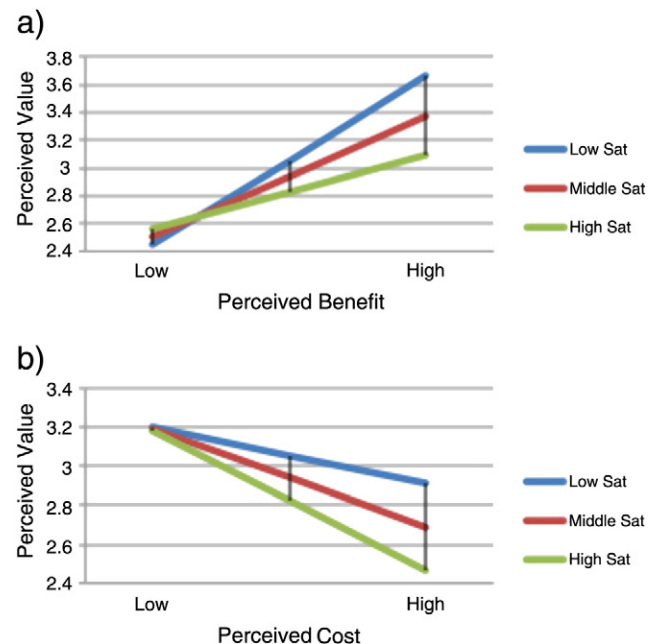


Fig. 3. a. Moderating effect of satisfaction on SB to PSV. b. Moderating effect of satisfaction on SC to PSV.

On the other hand, the R-square values for three different groups are 0.001 for low satisfaction, 0.126 for medium satisfaction, and 0.143 for high satisfaction. This implies that perceived cost generates more effects on perceive value of switching when the level of satisfaction is high. As shown in Fig. 3b, dissatisfied individuals are less sensitive to switching cost. The flat slope for dissatisfied customers implies that their perception of the value of switching does not change significantly as the switching cost changes. However, those highly satisfied with their current platform perceived less and less switching value as the switching cost increased. This implies that, for those unsatisfied customers, switching to other platform is always valuable, regardless of the level of cost or sacrifice.

In sum, it is reasonable to conclude that when determining switching value (1) highly satisfied customers are less attracted to the benefits provided by the alternatives and are very sensitive to the cost of switching, and (2) highly unsatisfied customers are more attracted to the benefits provided by the alternatives and are insensitive to the cost of switching.

4.3. Limitations and future research

Although our study contributes to the literature in various ways, because of the data collection and model construction design, several limitations are worth noting. First, this is a cross-sectional analysis and includes only intention as the final dependent variable. However, it is reasonable to believe that actual switching behaviors may serve as a better outcome variable to increase the internal validity of the research model and avoid common method bias. Hence future studies are encouraged to include objective behavioral style data to validate the research model. In addition, another significant limitation of this study is not taking dynamic effects into consideration. For example, service provider may reduce price or offer free traffic to retain customers with high switching intention [55]. While customers switch service providers to have their unfulfilled needs fulfilled, the necessity for them to leave is low if service providers can compensate those customers with various offers. Therefore, further studies are encouraged to conduct longitudinal study and incorporate dynamic effects in their models. Second, the target sample of this study is smartphone users. Switching intention may be affected by other attributes of mobile services as well, e.g., reliability and accessibility of connection. Lastly, only one context was used to test the proposed model. It would be useful to replicate this study across different types of innovative services to verify the robustness of the found results. Further research can also verify whether the same phenomenon can be observed consistently in different services.

4.4. Implications

In spite of above limitations, this study also generates several noticeable implications for practitioners and academia. First, this study contributes to the value-based decision research stream by showing the effect of affective components. In addition to perceived value, affective components (such as satisfaction) also contribute to switching decisions. Furthermore, this study also illustrates two roles that affective components may play. This implies that future value-based decision studies should take the effects of affective components into consideration. Based on the found results, researchers are encouraged to further explore the effect of other affective components in the switching decision-making process.

Second, this study contributes to traditional switching studies by showing that, in addition to reducing switching intention, satisfaction also changes customers' sensitivity to benefits and costs. The direct effect from satisfaction to switching intention has been validated by many studies. For example, pull-push-mooring model based switching studies e.g. [12] and continuance based studies e.g. [34] have specified the importance of satisfaction to the formation of switching or continuance intentions. With an understanding of the moderating effect of

satisfaction, this study contributes to the switching research stream by showing that satisfaction also plays a role in changing the magnitude of the effect of benefits and costs on perceived switching value. According to the results, highly satisfied individuals are less sensitive to benefits and more sensitive to costs when determining value.

While examining the interaction effect of satisfaction and other factors, past studies used mainly intention as the dependent variable. For example, Jones et al. [34] examined the interaction effect between satisfaction and switching cost (and the availability of alternatives) on continuance intention. Jones et al. [34] found that although unsatisfied individuals exhibit lower levels of continuance intention in general, they are more likely to make a repurchase decision when better choices are absent. This implies that unsatisfied individuals are forced to stay because they have no choice. In this study, I moved further and found that unsatisfied individuals are more sensitive to switching benefits. When alternatives are available and better service can be accessed, individuals with low levels of satisfaction tend to enlarge the effect of switching benefits and believe that switching is valuable, in contrast to highly satisfied customers. In addition, since unsatisfied individuals tend to attribute more weight to benefits offered by other providers when determining value, they are more likely to switch, given the positive correlation between perceived value and intention.

However, on the interaction between switching cost and satisfaction, the observed results were different from those found by Jones et al. [34]. The previous study found that switching cost generates an effect only when the satisfaction level is low, since highly satisfied individuals strongly intend to stay with their current service provider. Because of the high switching cost, unsatisfied individuals are forced to stick with their current service provider to avoid the extra costs resulting from switching. In this study, I found that switching cost generates more of an effect when the level of satisfaction is high. Unsatisfied individuals are less sensitive to costs because they are eager to leave their current service provider and, in contrast, highly satisfied individuals are very sensitive to any additional cost since they are not motivated to leave. Therefore, even while facing high switching costs, unsatisfied individuals still have a stronger perception that discontinuance is a wise or valuable decision.

The results of this study and previous studies indicate that (1) highly satisfied customers are not motivated to leave their current platform and, therefore, even slight cost for switching is considered as extra and unwanted. Therefore, they are very sensitive to switching cost when determining the value of switching. As an outcome, they tend to believe that switching is not a wise decision and, therefore, tend not to leave. (2) Dissatisfied customers are motivated to leave their current platform. Even though they may not actually switch since the switching cost is high, they still believe that leaving their current platform is a wise decision.

For practitioners, three major implications can be drawn from this study. *First*, this study provides more evidence to support how important it is for practitioners (e.g., service providers or platform owners) to maintain their customers' level of satisfaction. Satisfied customers are insensitive to the benefits of switching and are more sensitive to the cost of switching. One critical factor influencing satisfaction is that the service or platform provided must meet customers' expectations. Therefore, practitioners should pay attention to discovering customers' needs and improving or maintaining service quality in order to ensure the proper level of satisfaction. *Second*, the perceived benefit of switching was found to have a stronger effect than perceived cost. In addition, perceived benefit increases switching intention directly and indirectly through perceived value. Therefore, this also highlights how important it is for practitioners to persuade customers that sufficient benefits are offered.

Third, although perceived benefit has a positive effect on value, and perceived cost has negative effect, unsatisfied customers are more sensitive to benefits and less sensitive to cost. This implies that practitioners who attempt to attract unsatisfied customers from competitors should

highlight the benefits they can provide. Practitioners should make a thorough comparison of different service providers or platform attributes and try to emphasize the benefits of their own service or platform. As long as customers focus on the benefits that can be received from the competitors, they tend to rate the value of switching as high. On the other hand, practitioners who aim at retaining current customers should also pay attention to customer satisfaction. For satisfied customers, even though their switching intention is low, emphasizing the cost of switching can further suppress their switching intention.

5. Conclusion

This study addressed the insufficient consideration of rational components while using a value-based decision model to understand behavior in a non-mandatory context. Furthermore, this study also investigated how affective components may affect the rational decision-making process. Specifically, it is proposed that satisfaction, an affective component, not only suppresses switching intention but also changes individuals' sensitivity to the benefits and costs of switching, based on the needs-fulfillment concept proposed in needs theories. Data collected from 237 smartphone users confirmed most of the proposed hypotheses. The results showed that satisfaction has a negative impact on switching intention. In addition, satisfied individuals are more sensitive to the costs and less sensitive to the benefits of switching. In contrast, dissatisfied individuals are less sensitive to the costs and more sensitive to the benefits of switching. This interesting result generates several implications for academia and practitioners, as indicated above.

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